Marketing

Chapter 3: Product

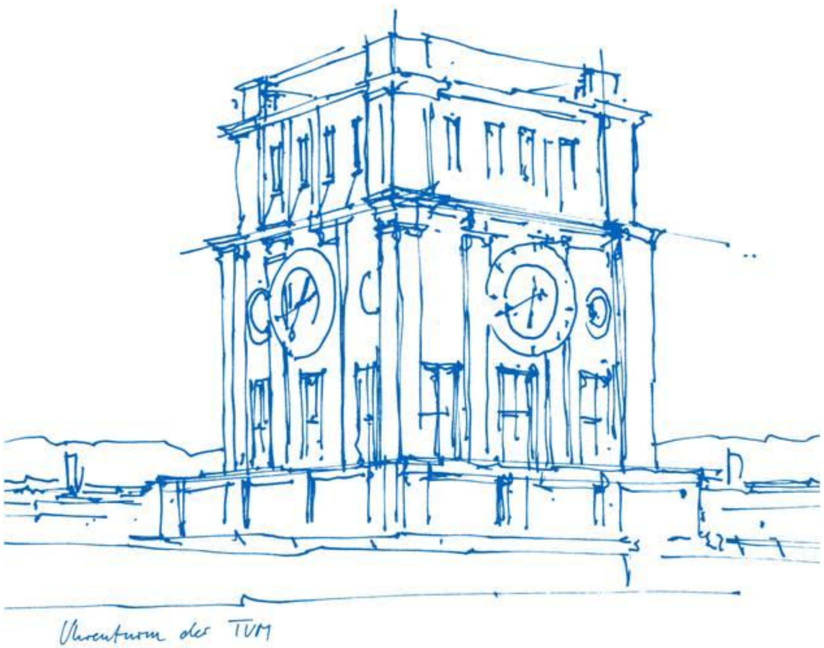
Prof. Dr. Martin Meißner

Technical University of Munich

TUM School of Management

Center for Digital Transformation

Professorship of Digital Marketing



Learning goals

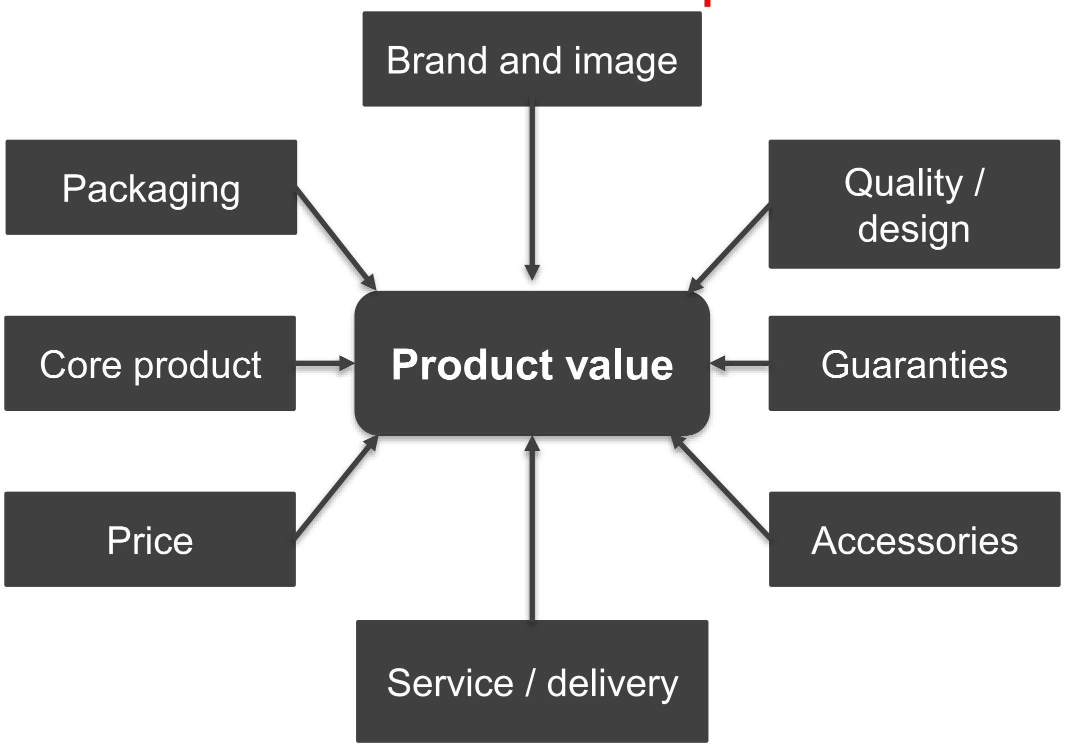
After studying this chapter, you will be able to:

- Explain the central components of a product
- Distinguish different approaches of product policy
- Discuss questions concerning the packaging of products
- Discuss of product mix policy tasks and decisions
- Learn about preference measurement with conjoint-analytic approaches

Product attributes

What is a product?
“[…] is a bundle of attributes […] that offer an observable benefit (of any kind) to its user” (Decker et al. 2015)

Chapter 2

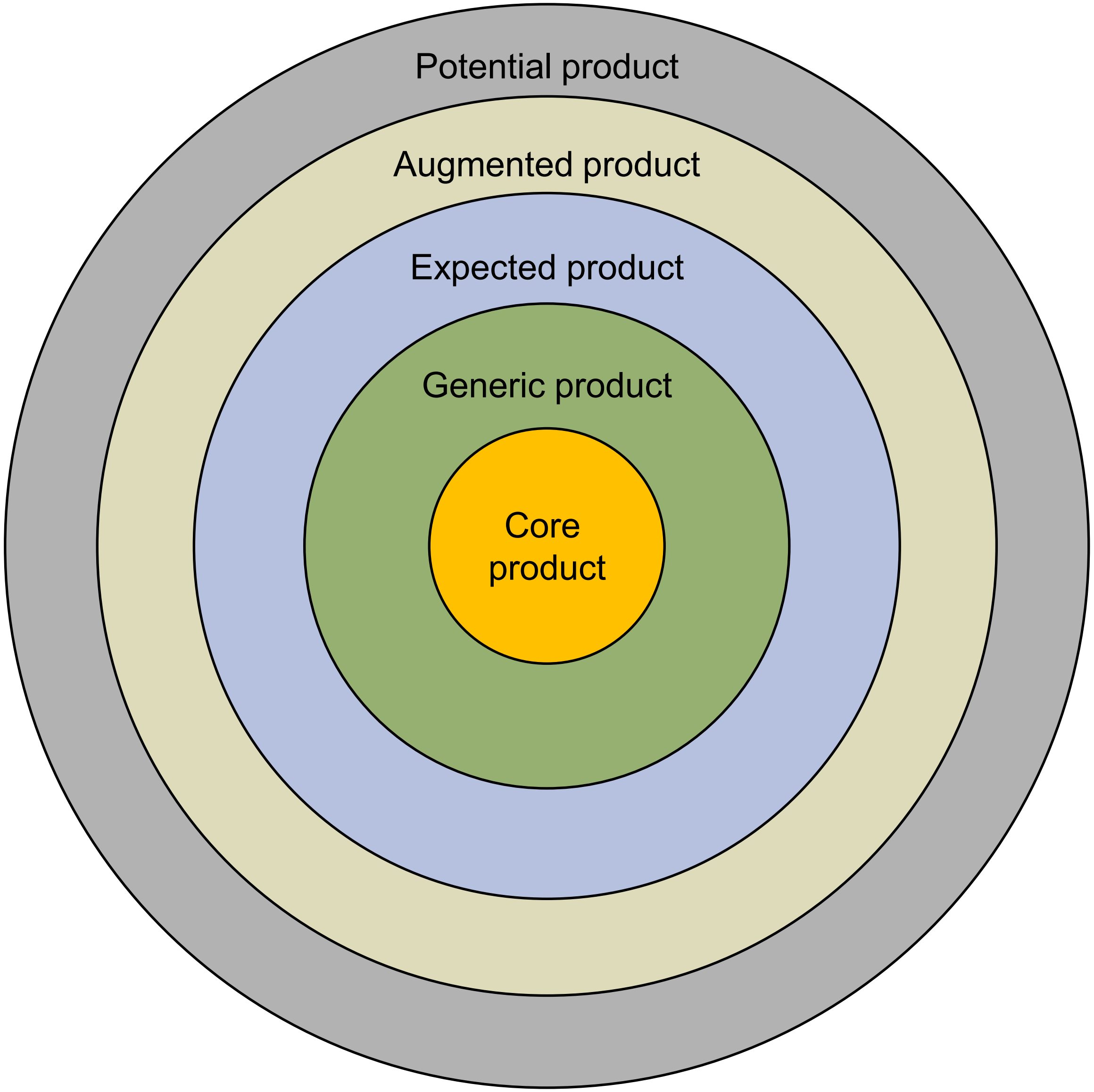


“The producer of drills for drill machines may believe that the customer needs a drill machine. What the customer really wants is a hole in the wall.” (Kotler et al. 2016, S. 43)

Product attributes

Hotel

Core product = rest and sleep
Generic product = hotel room includes a bed, bathroom, towels, dresser, a closet, etc.
Expected product = clean bed and bathroom, fresh towels, etc. (must-have features)
Augmented product = features that exceed customer expectations (e.g. high service level, free Wifi, etc.)
Potential product = how the product may evolve in the future



Product attributes

Level	Air Conditioner
1. Core Benefit	Cooling and comfort.
2. Generic Product	Sufficient cooling capacity (Btu per hour), an acceptable energy efficiency rating, adequate air intakes and exhausts, and so on.
3. Expected Product	*Consumer Reports* states that for a typical large air conditioner, consumers should expect at least two cooling speeds, expandable plastic side panels, adjustable louvers, removable air filter, vent for exhausting air, environmentally friendly R-410A refrigerant, power cord at least 60 inches long, one year parts-and-labor warranty on the entire unit, and a five-year parts-and-labor warranty on the refrigeration system.
4. Augmented Product	Optional features might include electric touch-pad controls, a display to show indoor and outdoor temperatures and the thermostat setting, an automatic mode to adjust fan speed based on the thermostat setting and room temperature, a toll-free 800 number for customer service, and so on.
5. Potential Product	Silently running, completely balanced throughout the room, and completely energy self-sufficient.

New technologies used to create experiential value

Hoyer et al. (2020) distinguish three dimensions of experiential value:

- **Cognitive value** is the experiential value that consumers receive as a result of processing the information and decision-making.
- **Sensory/emotional value** comprises the value consumers get from sensory stimulation and emotional attachment.
- **Social value** includes the value consumers receive by connecting to the social world around them because of the behaviors and relations that AI enables.



Hoyer, W. D., Kroschke, M., Schmitt, B., Kraume, K., & Shankar, V. (2020). Transforming the customer experience through new technologies. Journal of Interactive Marketing, 51(1), 57-71.

New technologies used to create experiential value

Table 3
A new framework for managing new technologies along the customer experience dimensions to create experiential value.

		Cognitive	Sensory/Emotional	Social
IoT	*O*	More information access	Object attachment	Make objects part of social system
	M	Data utilization	Anthromorphization	Human machine interfacing
AR/VR/MR	*O*	Catalyst for action	Complementing physical world	Create “possible worlds.”
	M	Appropriate visualization	Sensory stimulation	Customer immersion
Virtual assistant/Chatbot/Robot	*O*	Better decision	Making intelligence tangible	Integrating tech into peoples' lives
	M	Developing and improving algorithms	Humanized robotization	Humanization of interaction

O = Objective; M = Management task.

Your task:
Read pages 63 and 64 of the paper by Hoyer et al. (2020). Please come up with examples for how technologies influence customer experience dimensions.

Hoyer, W. D., Kroschke, M., Schmitt, B., Kraume, K., & Shankar, V. (2020). Transforming the customer experience through new technologies. Journal of Interactive Marketing, 51(1), 57-71.

Product – Digitalization

Two common strategies:

1) Transform **physical** products into **digital** products

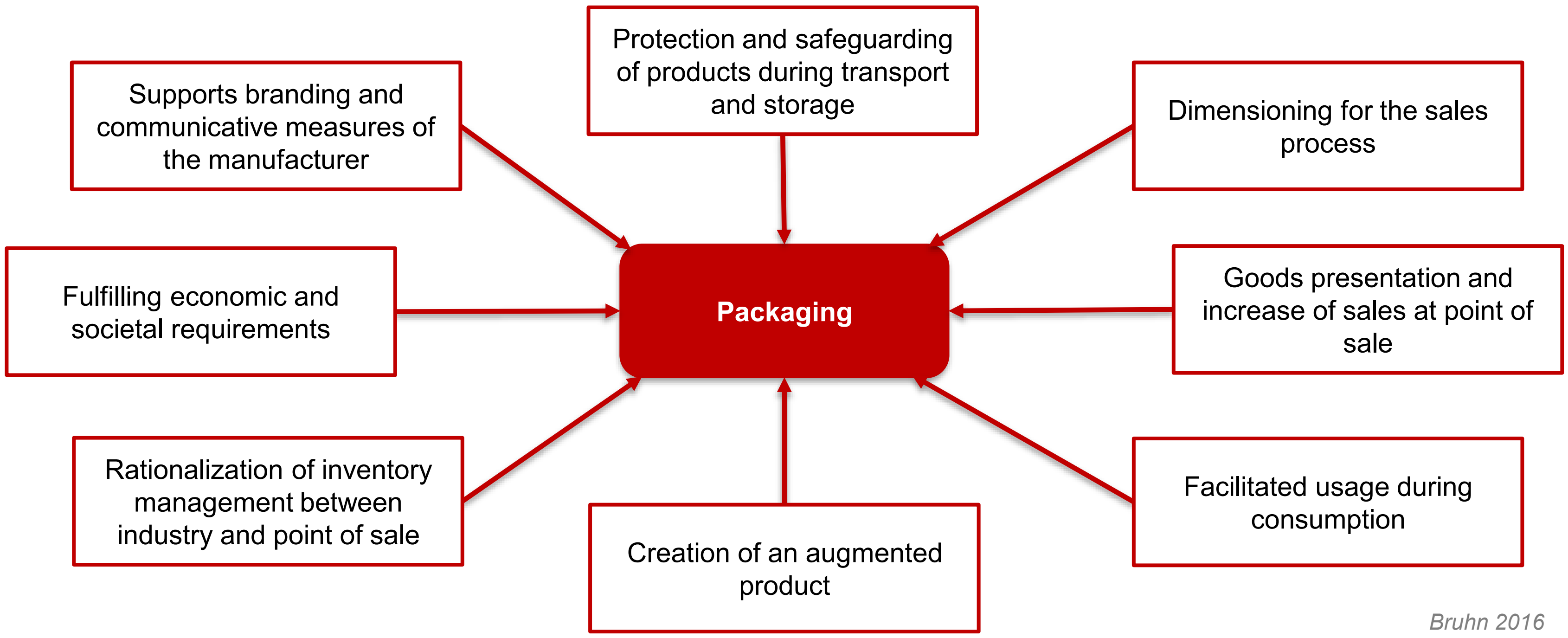
	Physical product	<u>Digital product</u>
Software	physical boxes, software on CD/DVD, printed manuals	available for download, PDF manuals and Youtube explanatory videos
Music	CDs, vinyl records	download, streaming
Books	printed books	eBooks
Movies/Videos	DVDs, Blue-ray	download, streaming

2) Augment **physical** products by **digital** elements (hybrid products)

<u>Physical product</u>	<u>Digital element</u>
Machines (B2B industries)	Sensors & software enable predictive maintenance and other new services such as optimization, consulting services etc.)
Cars	Entertainment system, driver assistance systems, …
Toothbrush	Sensors & software track brushing and keep kids entertained (e.g. Playbrush, Philips Sonicare kids)

Packaging

Decisions about packaging include all measures which are involved with enwrapping products (Bruhn 2016, S. 148)



Bruhn 2016

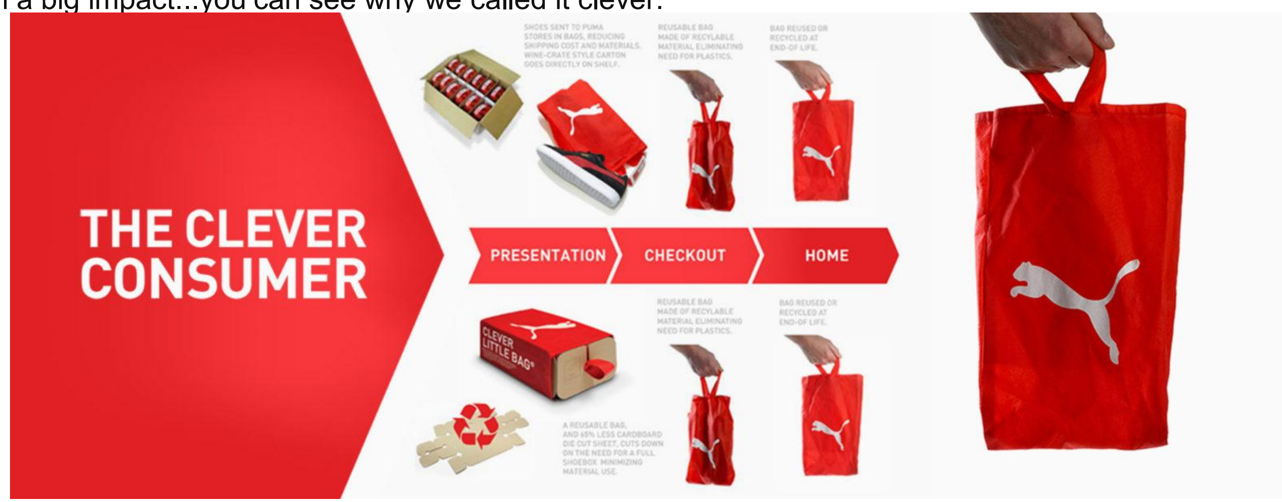
Packaging – Examples

Help! What does sustainable packaging mean?
What kind of packaging should my business use?

Packaging – Examples

Packaging – Examples

The cardboard structure is die cut from one flat piece of material and has no additional printing or assembly, thus it can be returned to the stream faster and more efficiently. The structure was created with four walls that taper in to allow for secured stacking, another important element left over from the original shoebox. The bag is non-woven which means less work and waste (it is stitched with heat). It protects the shoes from dust and dirt in the warehouse and during shipping. The “clever little bag” is an iconic brand element upon leaving the store as it replaces the plastic shopping bag, and can also be repurposed for creative reuse. The bag is made of non-woven polyester consisting of polypropylene, and eventually is also recyclable. With our ‘clever little bag’, Puma kicks-off the next pivotal phase of it’s sustainability program. The tens of millions of shoes shipped in our bag will reduce water, energy and diesel consumption on the manufacturing level alone by more than 60% per year. In other words: approximately 8,500 tons less paper consumed, 20 million Mega joules of electricity saved, 1 million liters less fuel oil used and 1 million liters of water conserved. During transport 500,000 liters of diesel is saved and lastly, by replacing traditional shopping bags the difference in weight will save almost 275 tons of plastic. That such a little bag can have such a big impact...you can see why we called it clever.

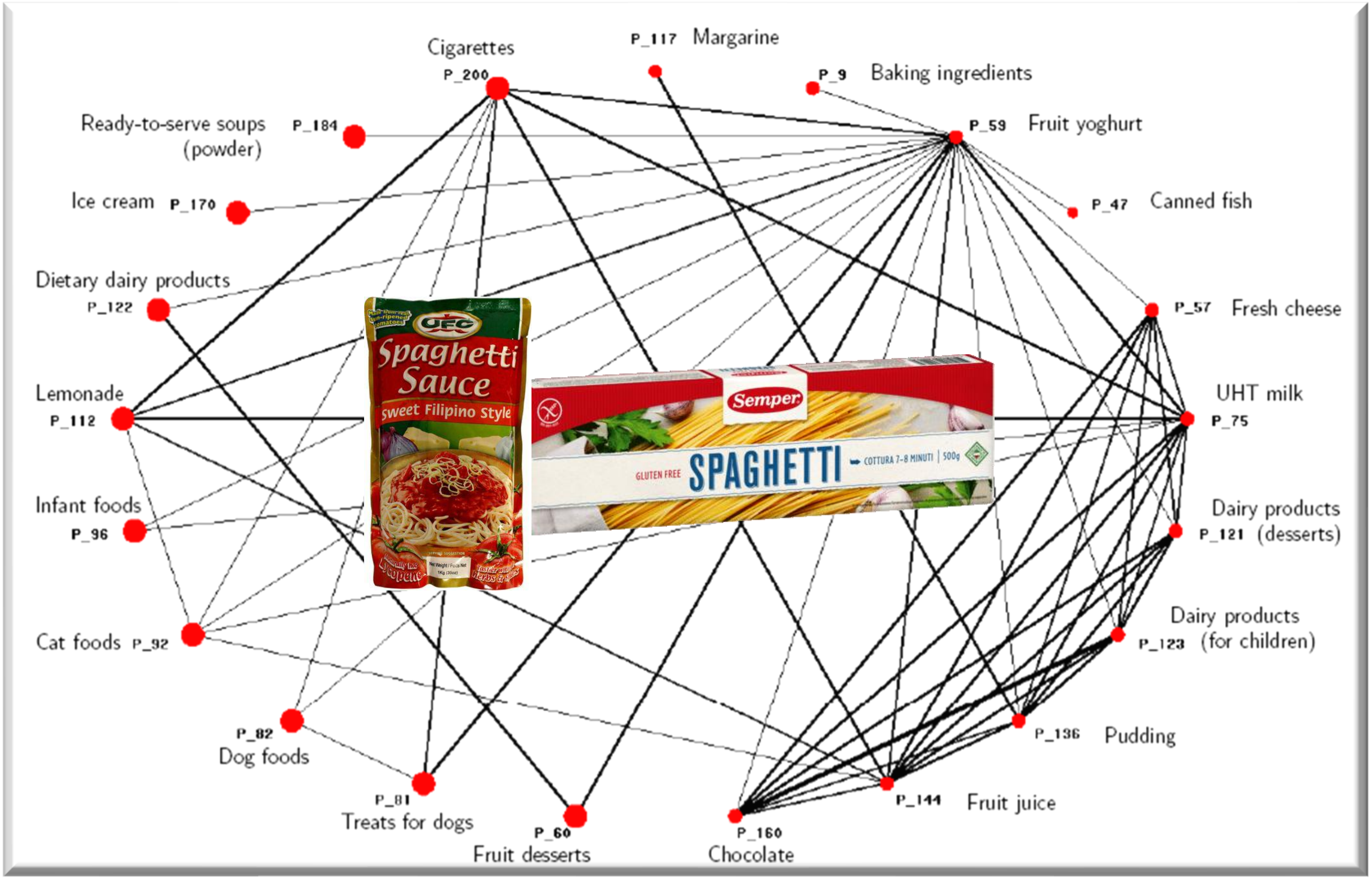


https://fuseproject.com/work/puma-clever-little-bag

Product assortment decisions

	Producer	Point of Sale
Product mix breadth	**Number of product lines** (e.g., chemical group with product lines for detergent, cosmetics, glue, and industrial coating)	**Number of product groups** (e.g., department store with product groups groceries, household hardware, shoes and textiles)
Product mix depth	**Number of articles in a product line** (e.g., product line for detergent with the products light- & heavy duty, wool, for color and fabric softener)	**Number of article groups in a product group** (e.g., product group „textiles“ with article groups for women, men and children)

Product assortment decisions



Product elimination

Possible causes:

- Continuously shrinking market.
- Increasing advantages of competitors.
- New legal requirements that cannot or will not be met.
- Declining sales, market share, contribution margins, profitability, etc.
- Reduction of production capacities.
- Negative test results/headlines.
- Excessive customer service costs.
- Scarce resources.

Possible consequences and risks:

- **Fixed cost coverage**
- **Image effect**: Damage to image when eliminating an image carrier ("classic") or image gains when eliminating questionable products.
- **Purchasing alliance**: Elimination can impair profitable alliance.
- **Know-how gap**: elimination can be more costly than continuation of production.
- **Relationship with suppliers and customers**: Product elimination may adversely affect business relationships.

AI-based recommender systems

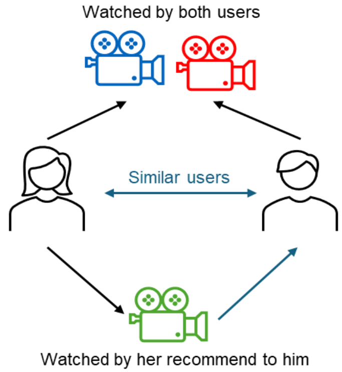
Collection
User, browsing history, social media or other sources.

Storage
Data storage, suitable database model and system

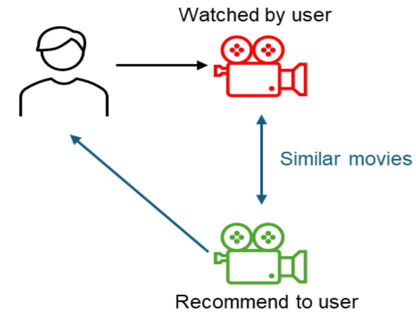
Analysis
Computational statistics and AI algorithms.

Filtering
Application of recommendation

Collaborative Filtering:



Content-Based Filtering:



Filtering Methods

AI-based recommender systems

Netflix Movie Thumbnail Customization Example

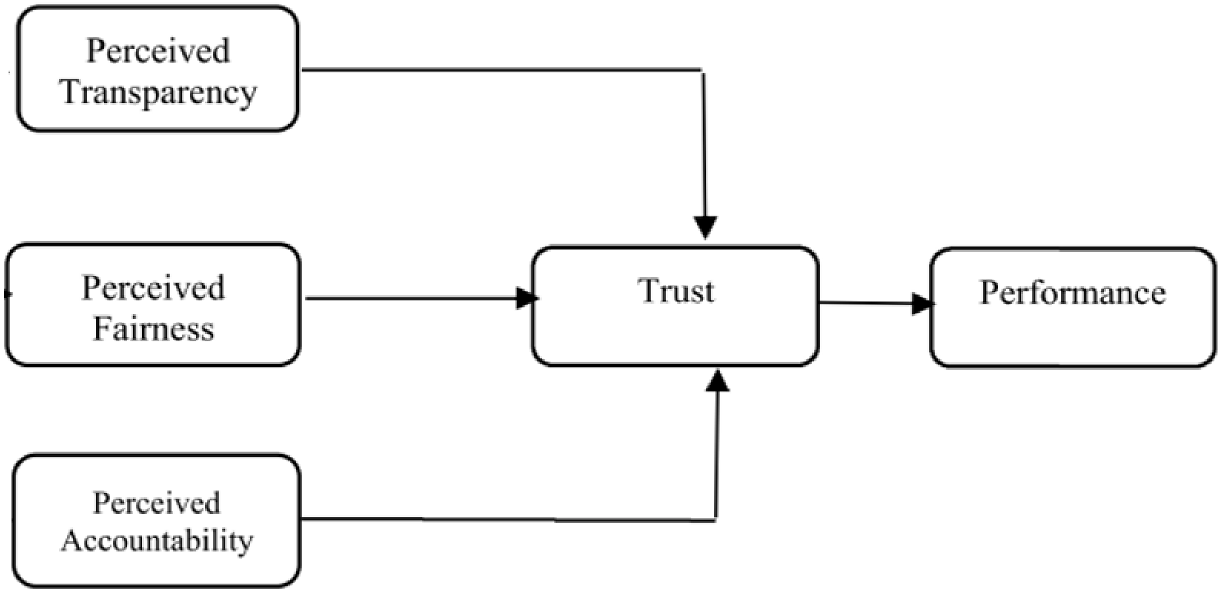
NETFLIX

Image retrieved from: https://netflixtechblog.com/artwork-personalization-c589f074ad76/

Consumer perception of AI-based recommender systems

FAT (sometimes FATE):

- **F**airness
- **A**ccountability
- **T**ransparency
- (**E**xplainability)



Relationships between Perceived Transparency, Fairness, Accountability, Trust and Performance (Shin 2021)

Consumer perception of AI-based recommender systems

Fairness: “Fairness in AI contexts refers to algorithmic processes that should not create discriminatory or unjust consequences (Shin 2021). A fairness question begins with concern that algorithms do not always behave objectively. The fairness in algorithms is rooted in the critical realization that algorithm systems can unfairly and systematically discriminate against certain people in favor of others” (Noble 2018).

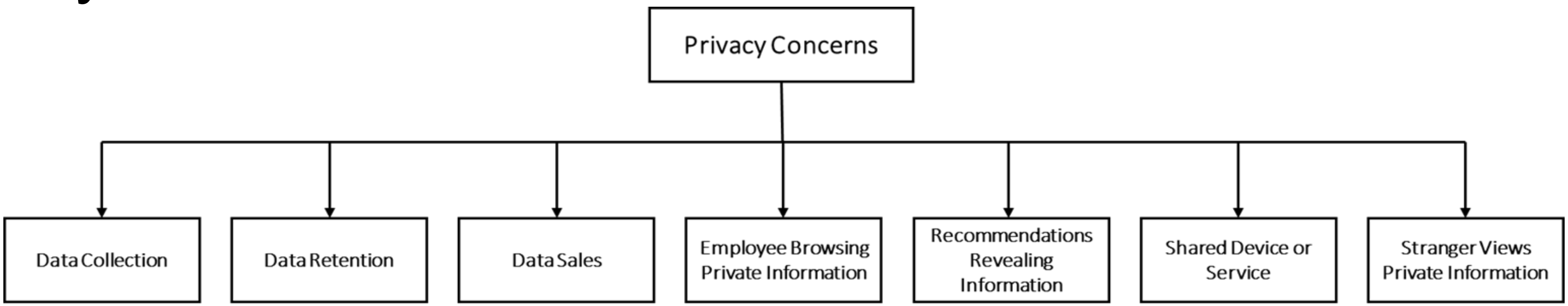
Consumer perception of AI-based recommender systems

Accountability: “Algorithmic accountability is the process of clarifying the culpability of underlying algorithms for harm when algorithmic decision-making results in undesirable consequences (Moller et al. 2018). It is a concept measure aimed at holding the providers of automated decision systems responsible for the results generated by their programmed decision-making” (Lewis et al. 2019)

Transparency: “The notion of transparency mandates that decisions by algorithmic filtering are clear and understandable to users” (Shin 2021).

Explainability:“Explainability is the extent to which the inner mechanism (black-box process) of an algorithm system can be explained in human languages (Shin 2021). It can be a seeing-through to comprehend the internal mechanism by which an algorithm operates” (Rai 2020).

Privacy concerns in regard to recommender systems



based on Jeckmans et al. 2013

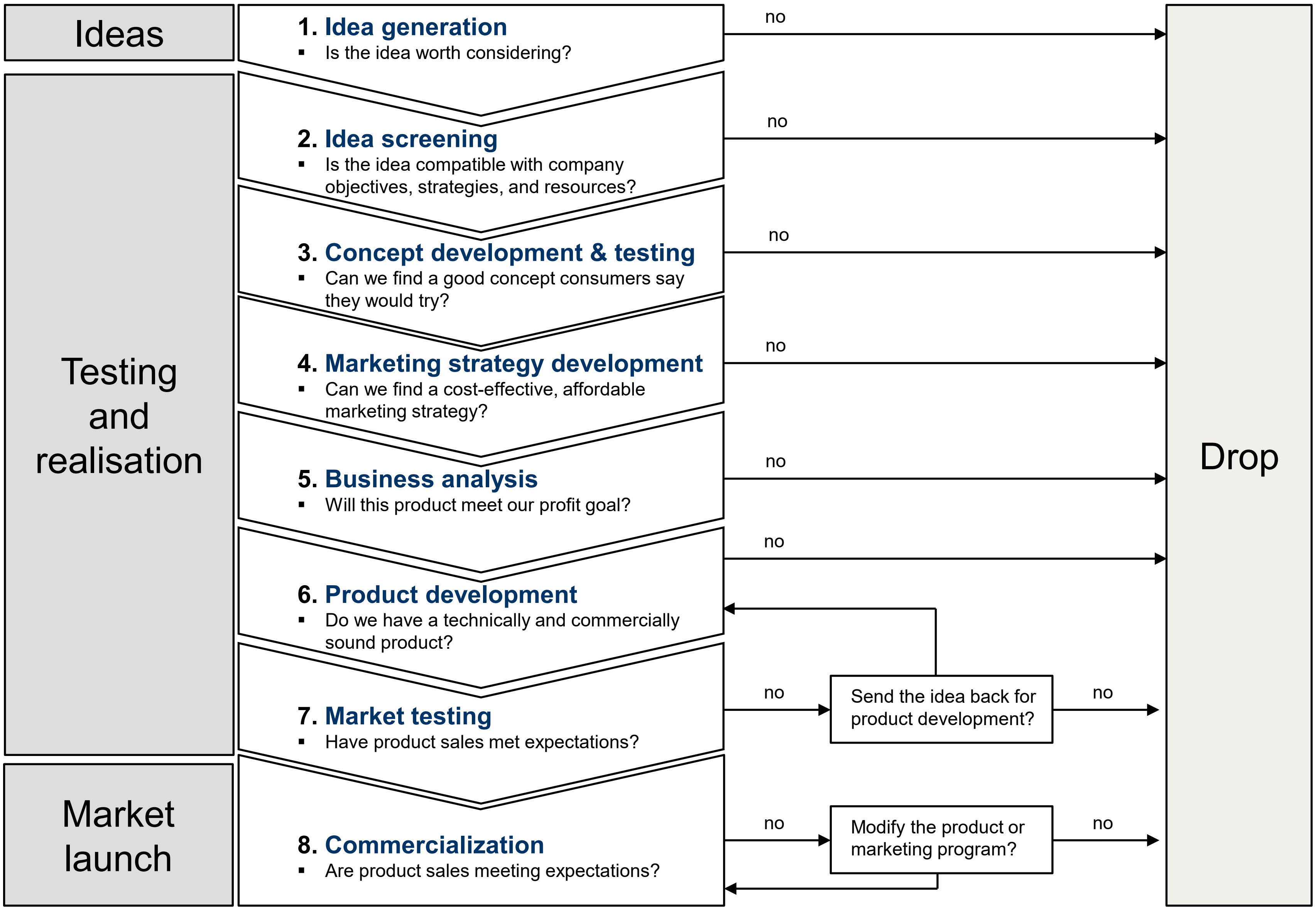
https://twitter.com/

Factors influencing privacy concerns

CREEPINESS	*NEED FOR PRIVACY*	*PERCEIVED RISK*
This feeling of creepiness is caused by emotional unease, ambivalence and it is caused by difference between an individual's expectations regard the technology’s capabilities and what it can really do in reality.	The need for privacy of a user reflects an individual's general concern about the confidentiality, use, or disclosure of private information by organizations. This sensitivity varies from person to person and is related to individual characteristics.	Perceived risk means the extent to which the use of systems using artificial intelligence services is perceived by users as a possible reason for data leakage. This applies in particular to sensitive data such as bank account numbers

Bouhia et al. 2022

Stages of product development

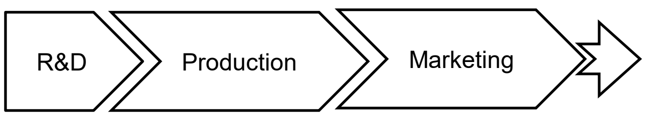


Kotler, Keller (2009)

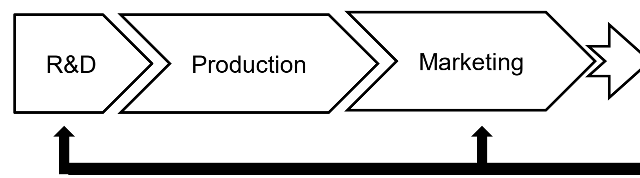
Sources of product innovation

Where do the ideas come from?

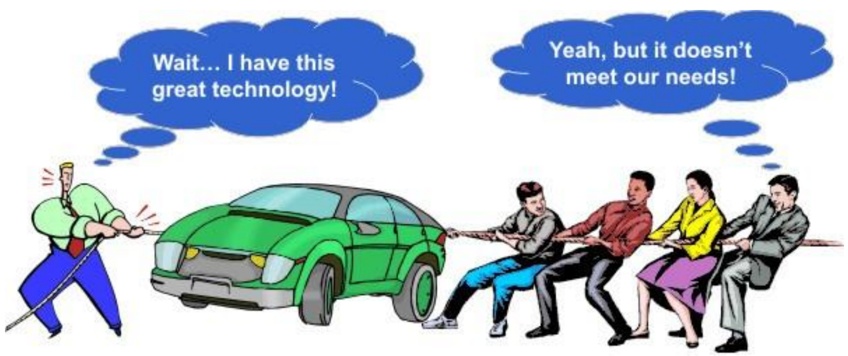
Engineers:
technology push



Customers:
market pull



"If I'd asked people what they wanted, they would have said 'a faster horse'."
(Henry Ford)
"Customers just don‘t know what‘s possible"
(Steve Jobs)



"Focus on the impact of having a problem solved… X Prize Lab takes a very 'market-pull'-oriented approach" (MIT X Price)
"Determine what your customers need and work backward, even if it requires learning new skills"
(Jeff Bezos)

“Is an entrepreneur with a market need in mind and a product to fit that need better positioned for success than an entrepreneur who starts with a technology but “needs” a need?”

Sources of product innovation

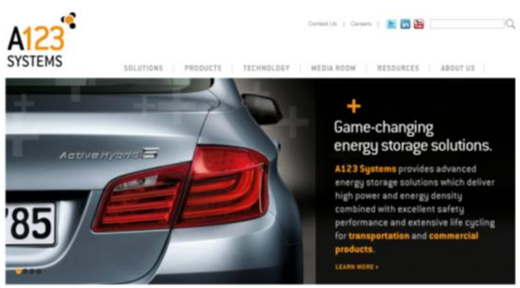
Marketing people about technical people	Technical people about marketing people
Have a very narrow view of the world	Want everything now
Never finish developing a product	Are focusing on customers who do not know what they want
Have no sense of time	Quick to make promises they cannot keep
Are interested only in technology	Cannot make up their minds
Do not care about costs	Cannot possibly understand technology
Have no idea of the real world	Are superficial
Are in a different world	Too quick in introducing new products
Always looking for standardisation	Want to ship products before they are ready
Should be kept away from customers	Are not interested in the scientist's problems

Sources of product innovation

Where do the ideas come from?

Engineers:
technology push

R&D → Production → Marketing → Need?



Customers:
market pull

R&D → Production → Marketing → Expressed market need

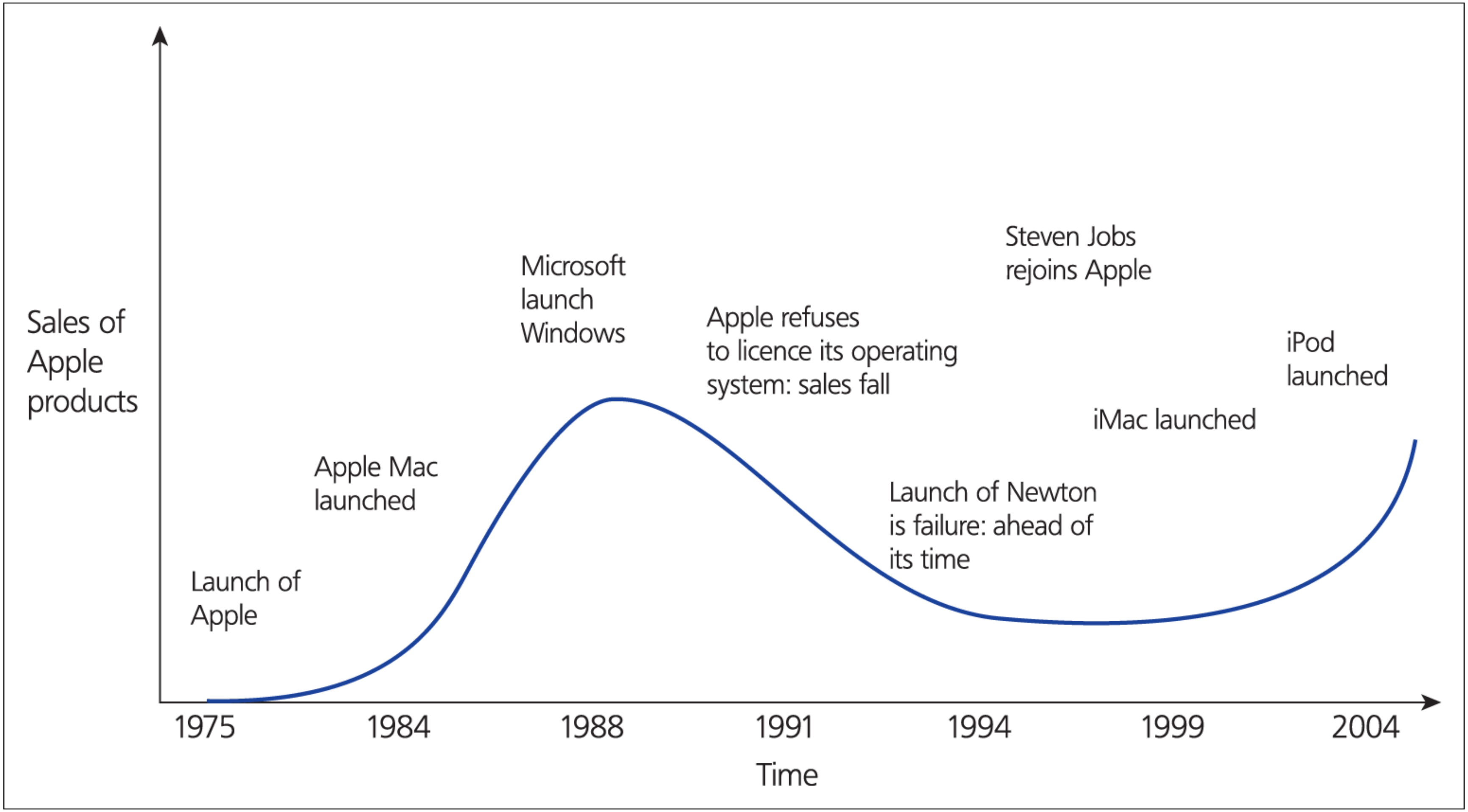
Gillette

double-blade razor (1976)

Mach 3, three-bladed razor (1998)

Gillette Fusion, five-bladed razor (2006)

The rise and fall and rise of Apple



Steve Jobs in the 80‘s

Steve Jobs 1997

Context variables

Context variable	Modifiers to the basic process
Sector	Different sectors have different priorities and characteristics – for example, scale-intensive, science-intensive
Size	Small firms differ in terms of access to resources, etc. and so need to develop more linkages
National systems of innovation	Different countries have more or less supportive contexts in terms of institutions, policies, etc.
Life cycle (of technology, industry, etc.)	Different stages in life-cycle emphasize different aspects of innovation - for example, new technology industries versus mature established firms

Context variable	Modifiers to the basic process
Degree of novelty-continuous vs. discontinuous innovation	'More of the same' improvement innovation requires different approaches to organization and management to more radical forms. At the limit firms may deploy 'dual structures' or even split or spin off in order to exploit opportunities
Role played by external agencies such as regulators	Some sectors -e.g., utilities, telecommunications and some public services - are heavily influenced by external regimes which shape the rate and direction of innovative activity. Others - like food or healthcare - may be highly regulated in certain directions

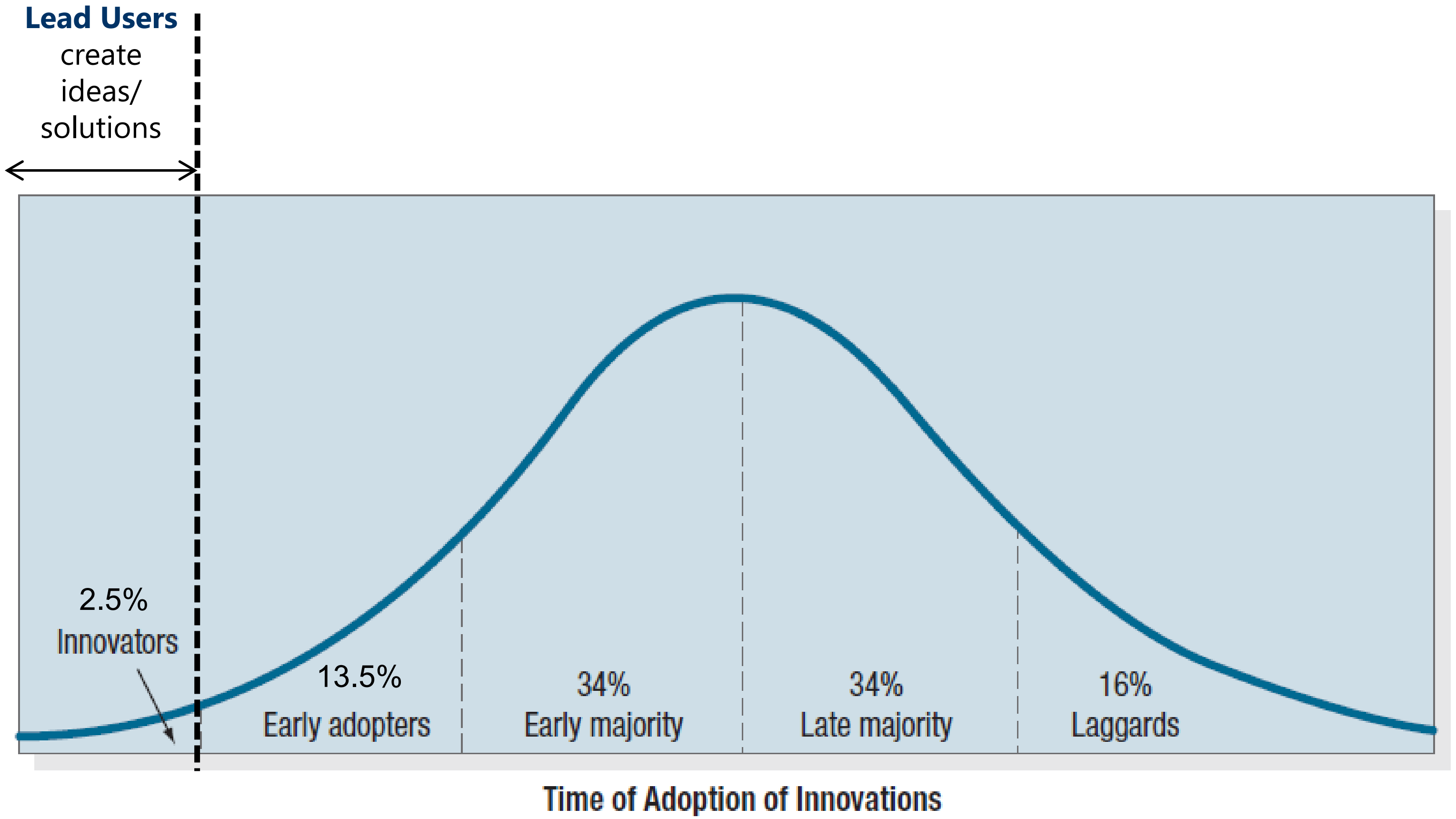
Lead users and idea generation

Lead users can be critical. These typically:

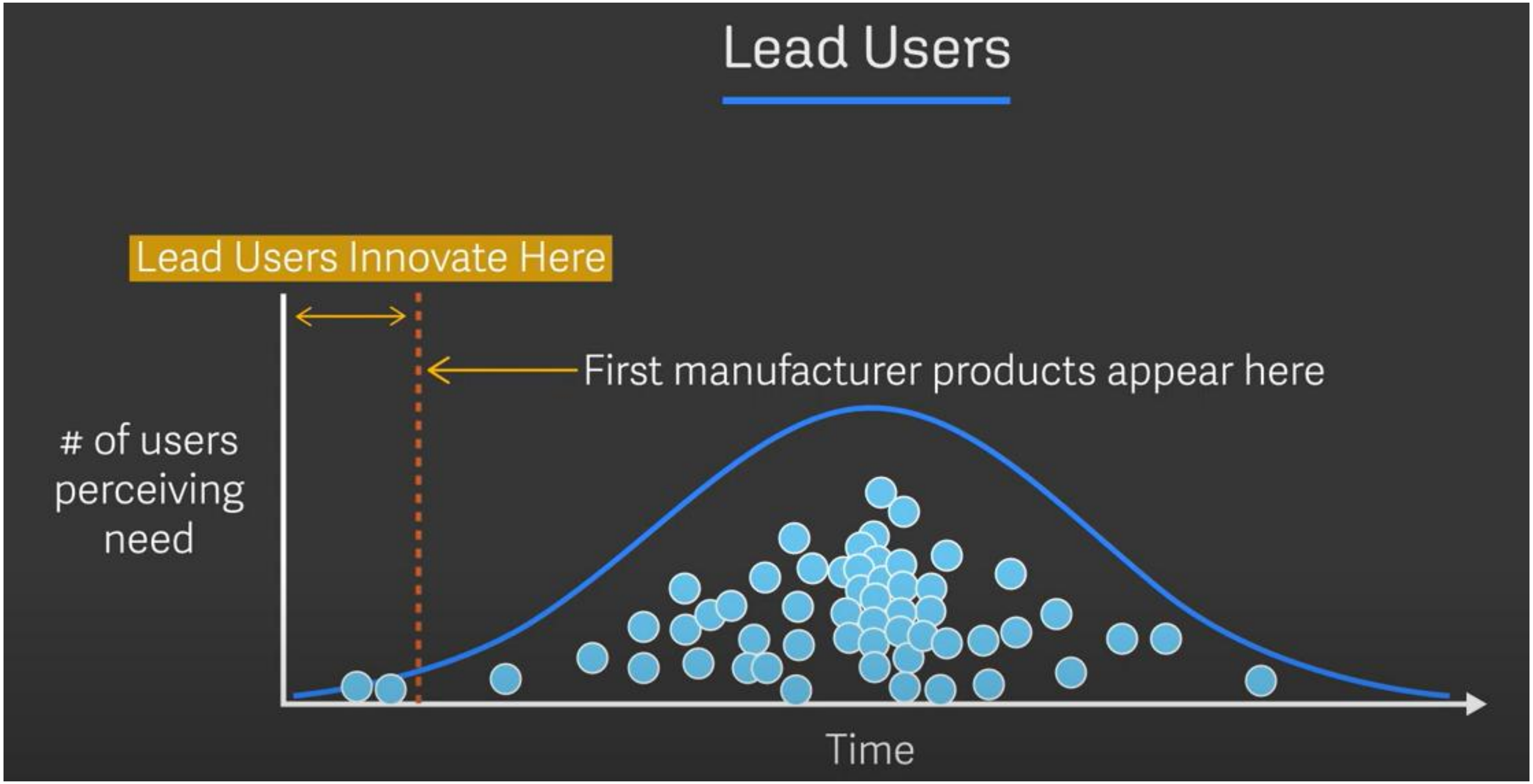
- **Recognize requirements early** - are ahead of the market in identifying and planning for new requirements.
- **Expect high level of benefits** - due to their market position and complementary assets.
- **Develop their own innovations and applications** - have sufficient sophistication to identify and capabilities to contribute to development of the innovation.
- **Perceived to be pioneering and innovative** - by themselves and their peer group.

Lead users and idea generation

Adopter categorization can also be used for new product development



Lead user integration



https://www.youtube.com/watch?v=bNXEwAxM41E

Lead user integration

Phase	Content
1 **Goal Generation and Team Formation**	• Firm provides an idea-generation-goal for the project and forms an LU project team • Teams: 3 - 5 experienced people from marketing and technical departments • Typical project duration: six months
2 **Trend Research**	• Identifying and deeply understanding important market and technical trends in the field of interest • Identifying and interviewing leading experts in the marketplace • Selecting an important trend or several trends as the central focus of further project work
3 **Lead User Pyramid Networking**	• Identify and learn from users at the leading edge of the important trends selected as a study focus • Identify lead users both in the target market and in other markets with similar needs. • Pyramid networking technique relies on the fact that people with a strong interest in a topic tend to know people more expert than themselves
4 **Lead User Workshop and Idea Improvement**	• Lead users work with company personnel to improve the preliminary concepts generated in Phase 3 • Typically, 10 to 15 people attend this workshop • The entire group evaluates the concepts in terms of technical feasibility, market appeal, and management priorities

Success of the lead user idea-generation process

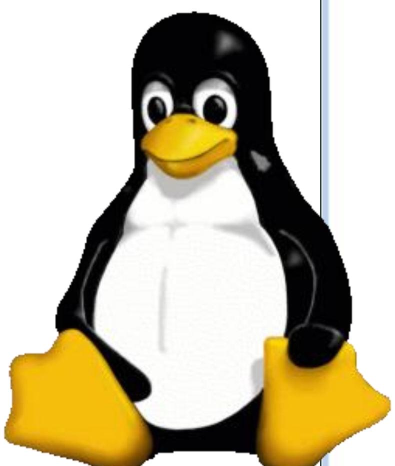
New products that have emerged from lead users' needs and solution data were **preferred** by potential users compared to product concepts generated by more traditional methods

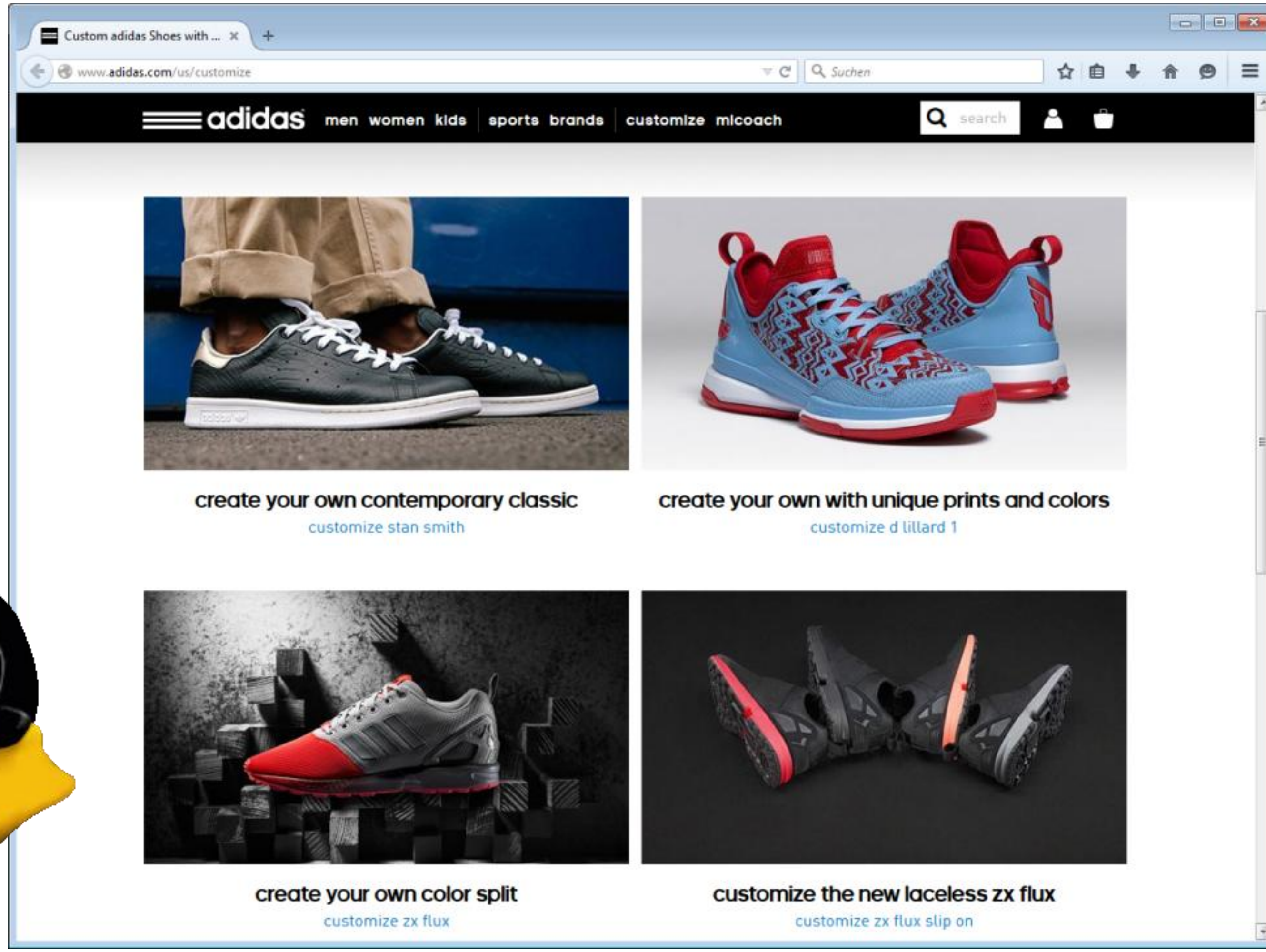
Idea generation at 3M with lead users (compared to non-lead users) led to **high-grade innovations**

The **commitment of lead users** tends to result in a fast and cheap innovation development

Source: according to Lilien et al. (2002), pp. 1045 et seq.; Trommsdorff /Steinhoff (2007), p. 278.

Customer co-creation



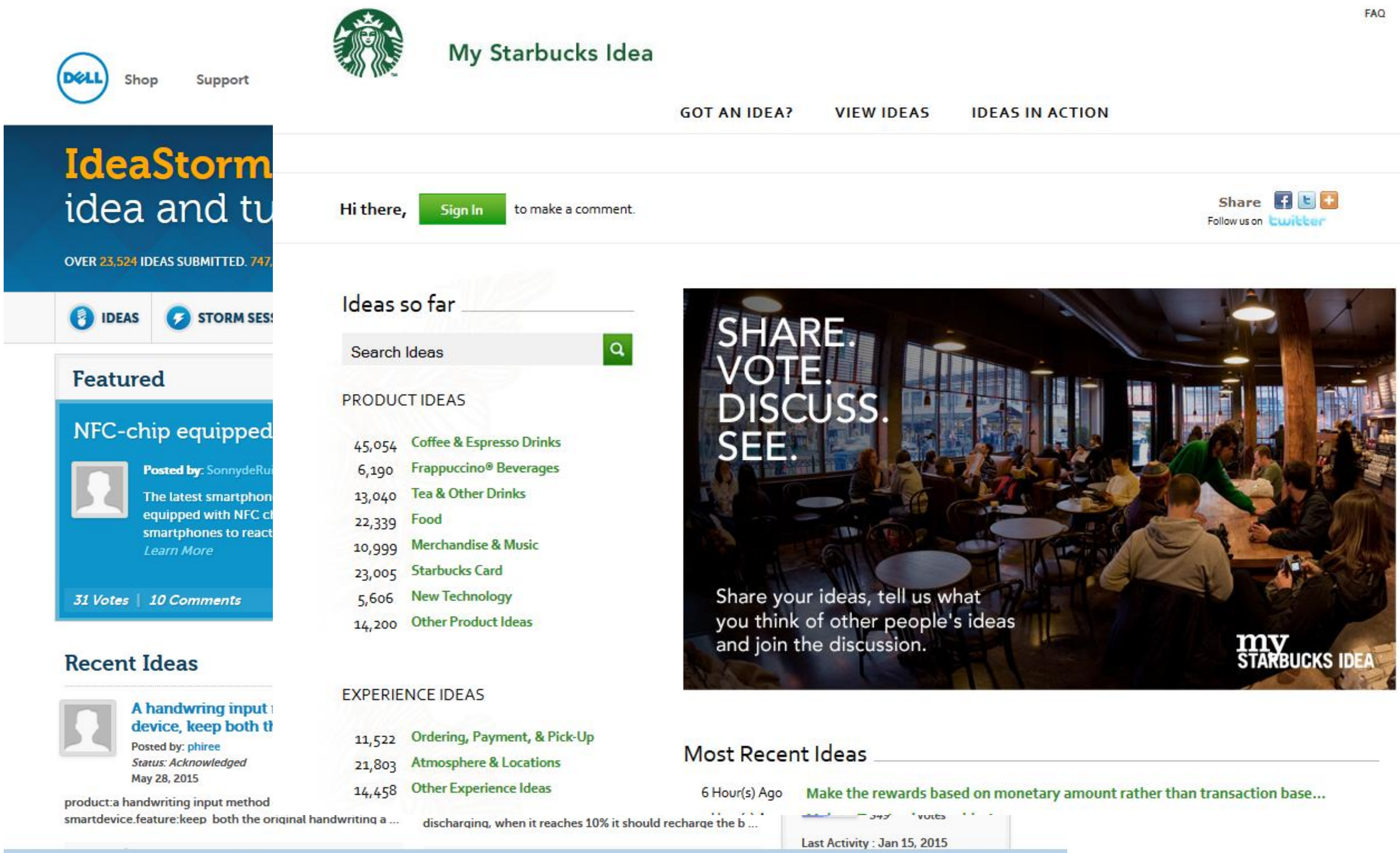


Customer co-creation

Benefits for the firm	Benefits for the customer
• Firm obtains access to need and solution information more easily • Innovation might better fit customers' needs → failure rates might decrease • Possibility to increase willingness to pay ("made it myself effect") • Activation of shared learning processes	• Material reward • Puzzling as being fun; considered as a challenge • Obtain acceptance from others • Connect to people with similar interests (via social networking sites)

Heidenreich S, Wittkowski K, Handrich M, Falk T. The dark side of customer co-creation: exploring the consequences of failed co-created services. *Journal of The Academy of Marketing Science.* May 2015;43(3):279-296

Customer co-creation – crowd-sourcing platforms



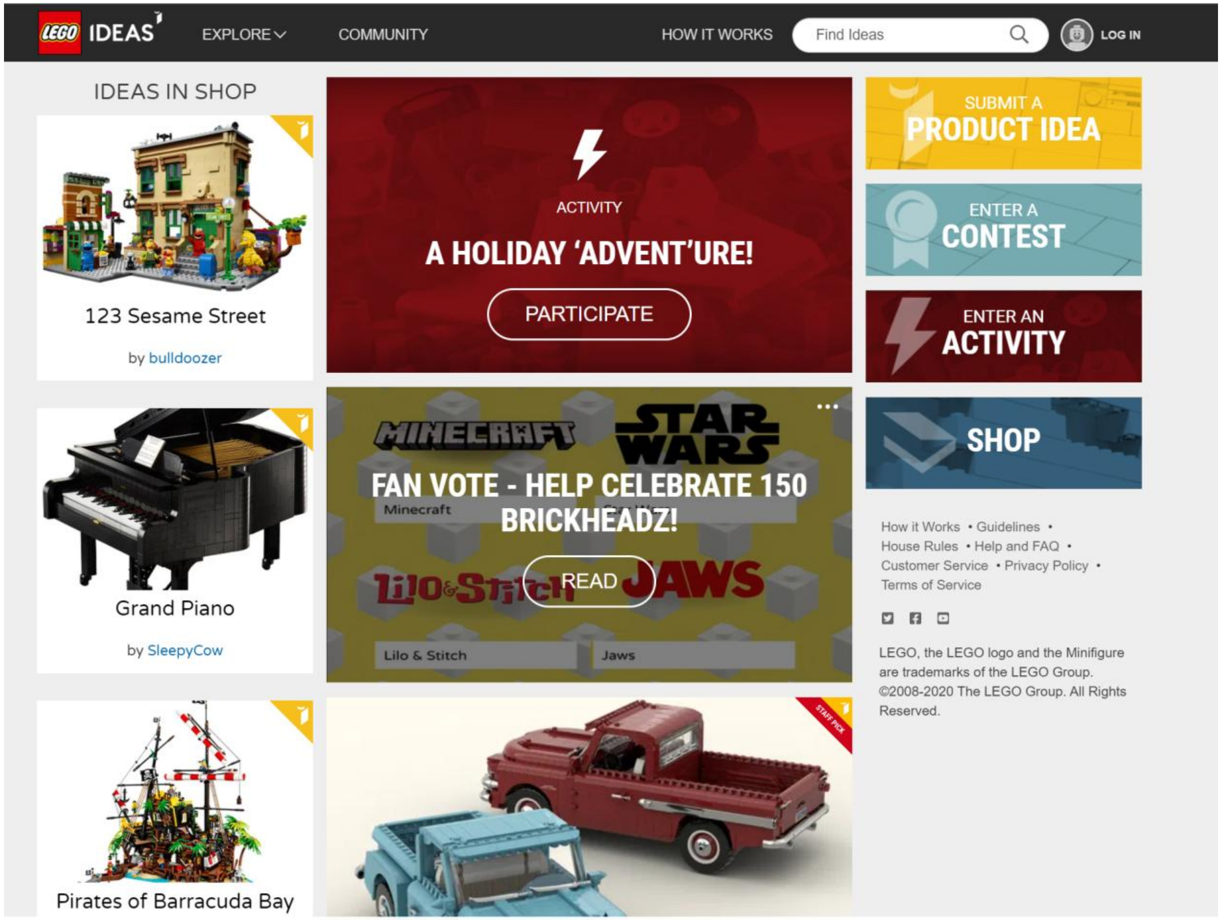
Customer co-creation

… but … why did/do so many customer co-creation projects fail?

- ☹ incremental innovation
- ☹ managing a huge amount of suggestions is not efficient
- ☹ expectation of the community is that they will see their products on the shelves [soon] after
  → declining passion for the brand
- ☹ different parties expect different things, and it's difficult to create fruitful dialog and, more importantly, to generate a joint outcome
- ☹ in the case of a (service) failure, customers using offers high on co-creation will be less satisfied than customers using the same service low on co-creation (due to higher disconfirmation of expectations)

Customer co-creation

… but … sometimes it works very well



Challenges in product development

- **High failure rates:**
  too much focus on technology push (failure rates: Germany ~65%, U.S. 70% to 90%)

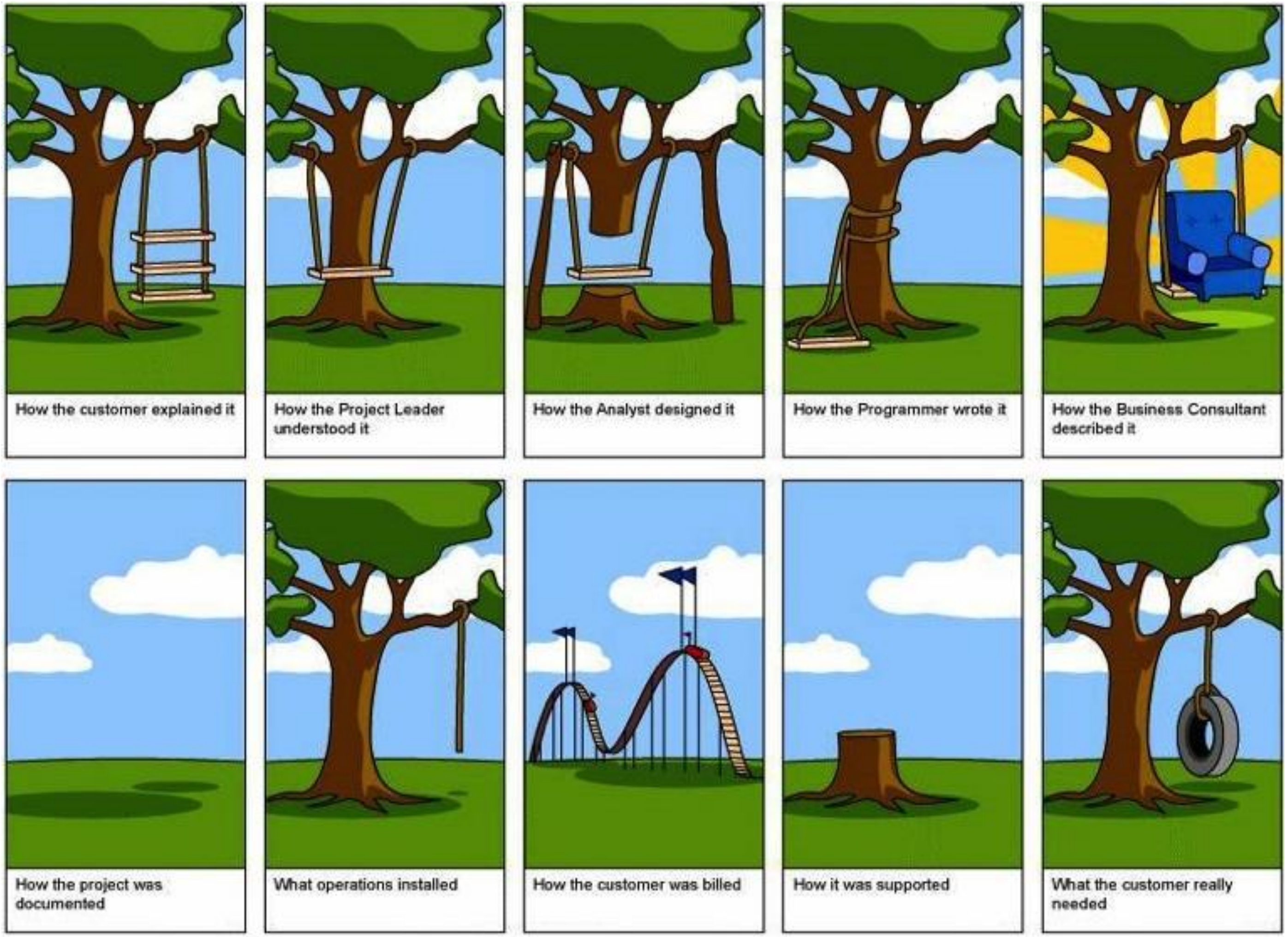
- **High resources expenditure:**
  high costs on R&D, market research, market introduction

- **Organizational resistance:**
  resistance against anything new at all levels of the company

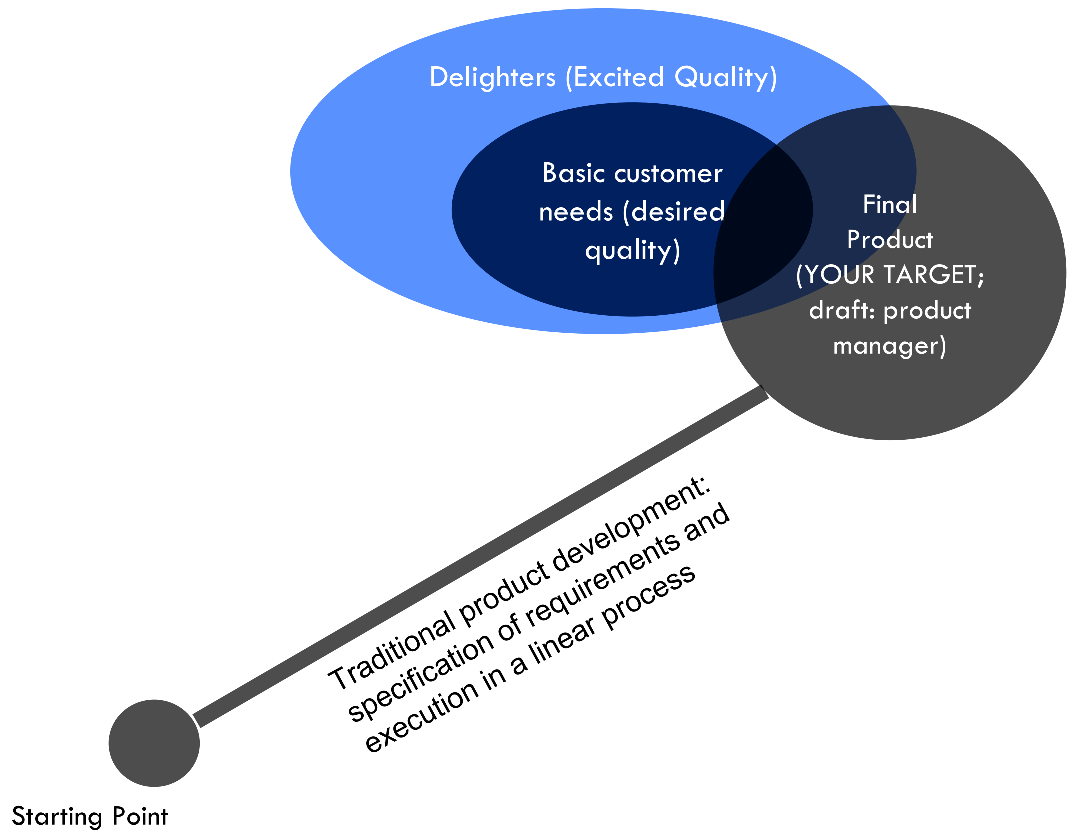
- **Market risks:**
  entering the market at the right time frame is critical… but when to enter?
  (too early = unaccepted product; too late = established market)

- **Complicated products:**
  Sometimes products become less simple to use when trying to develop innovatively
  (redundant/less helpful functionalities)

Challenges in product development

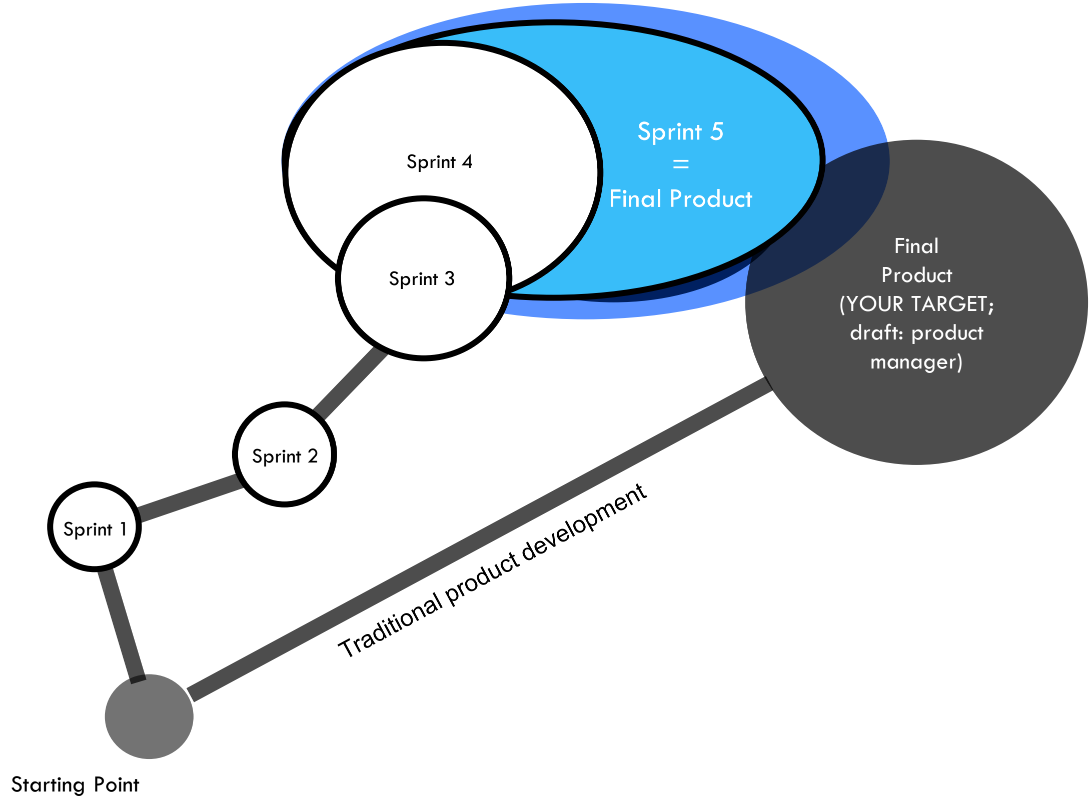


Agile product development / Lean startup method



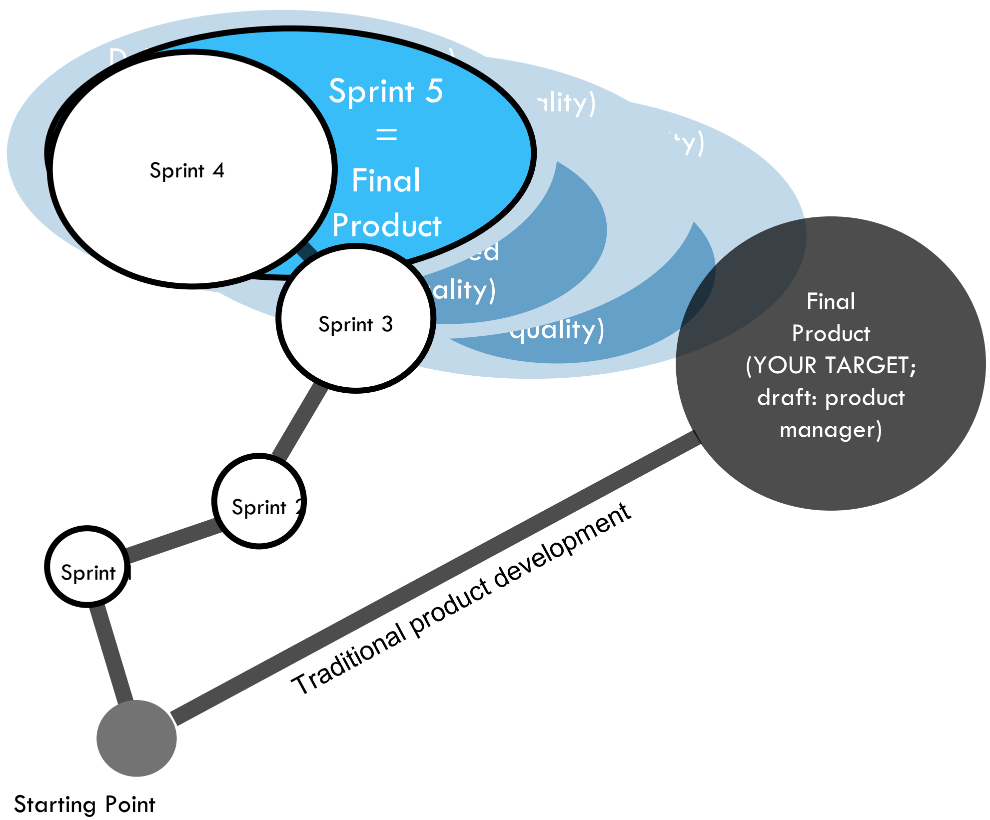
Agile product development / Lean startup method

Wouldn't it be better to divide your assumptions, to test them separately and to interact with your customer within the process?

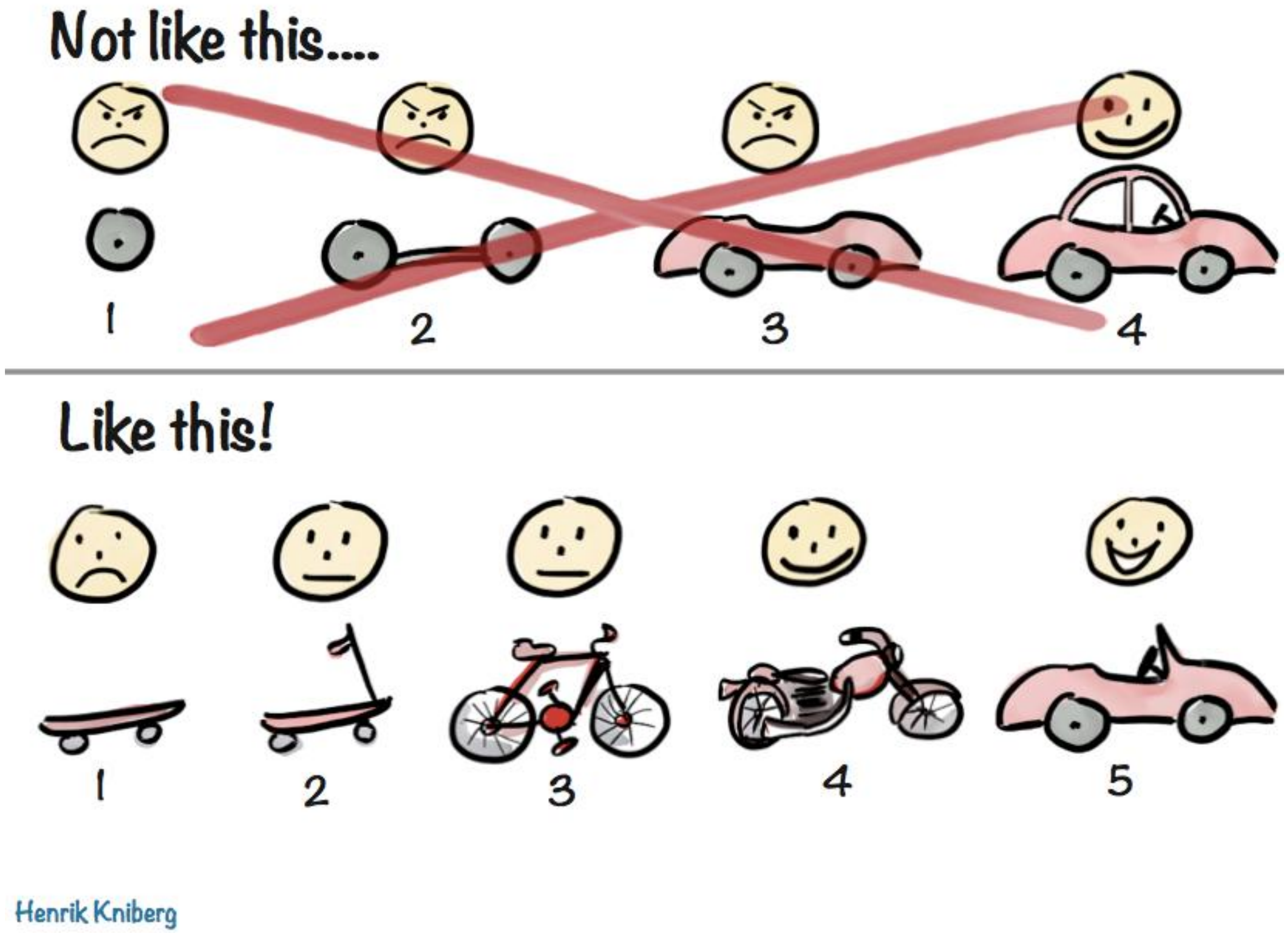


Agile product development / Lean startup method

…especially if your customer needs vary strongly over time?!



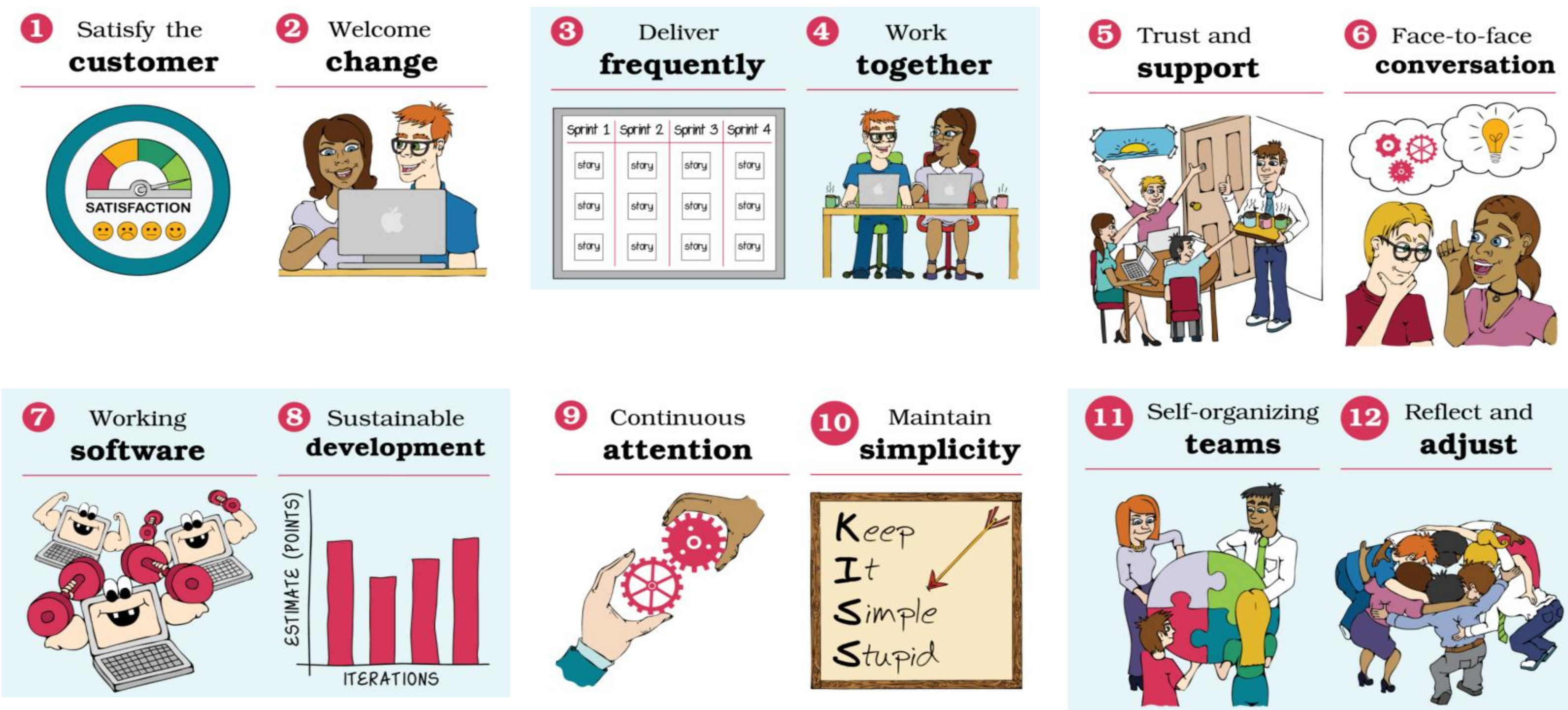
Agile product development / Lean startup method



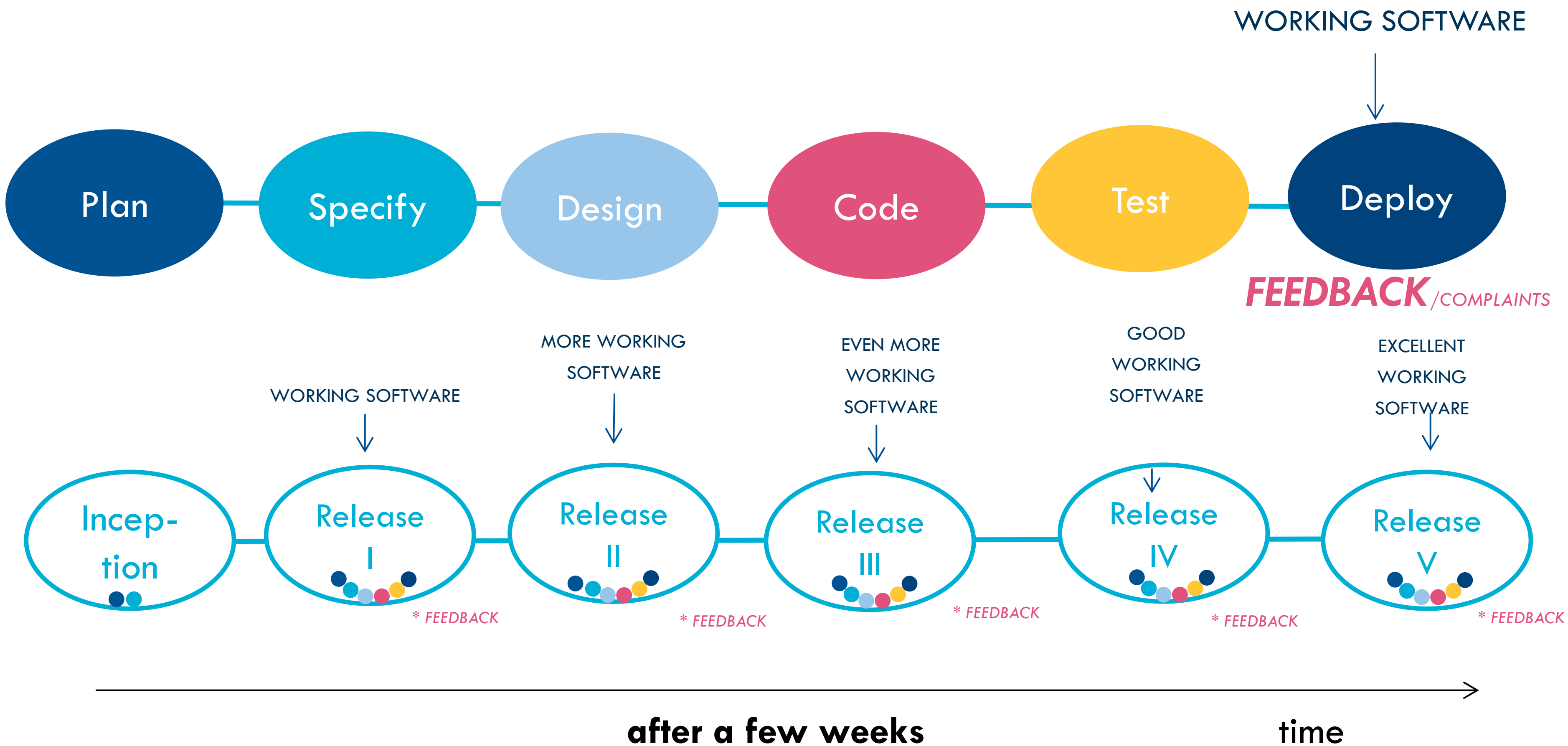
https://www.youtube.com/watch?v=0P7nCmIn7PM

Agile product development / Lean startup method

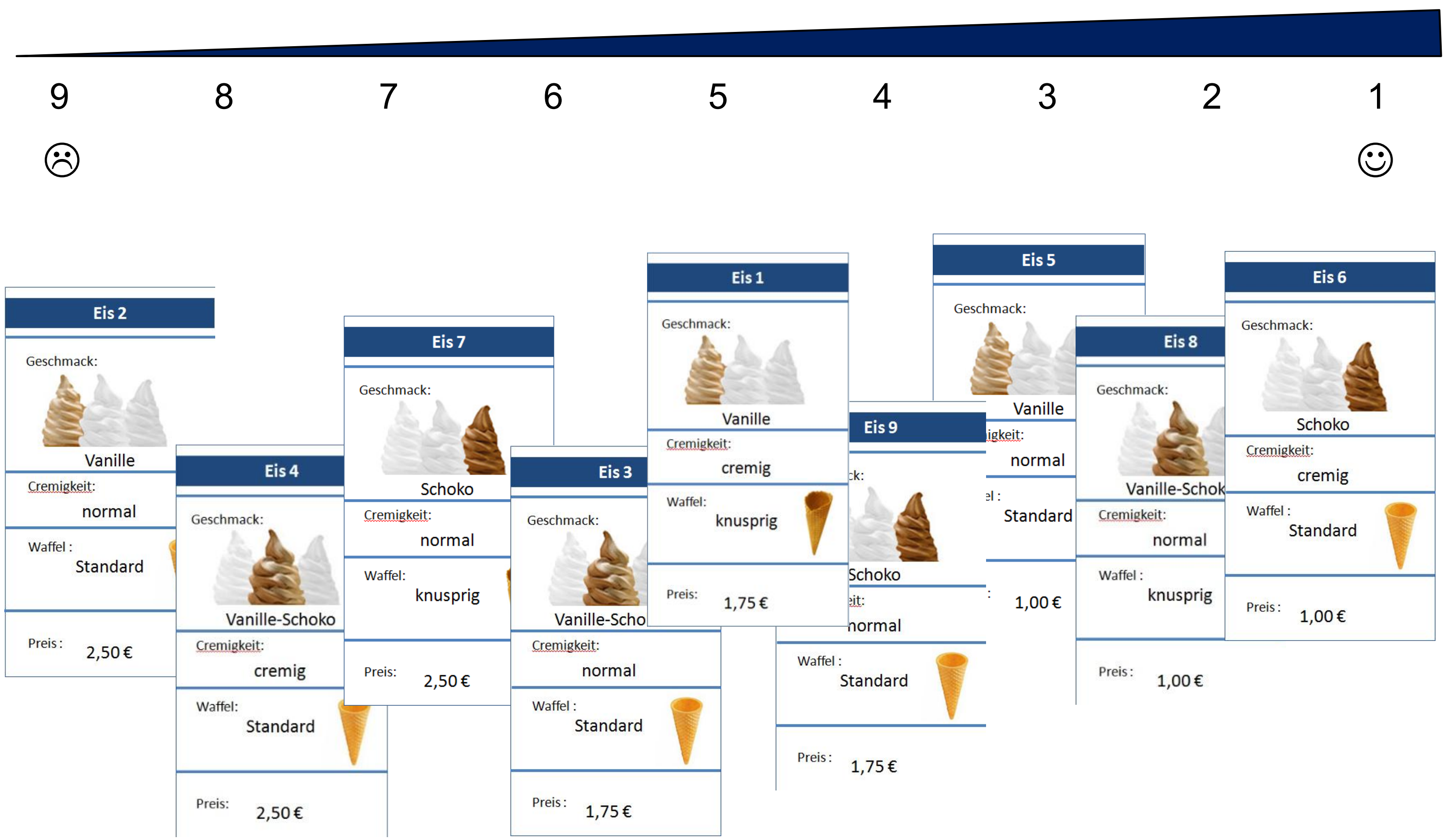
Agile is a mindset, not a methodology!



Waterfall vs. Agile: the example of software development



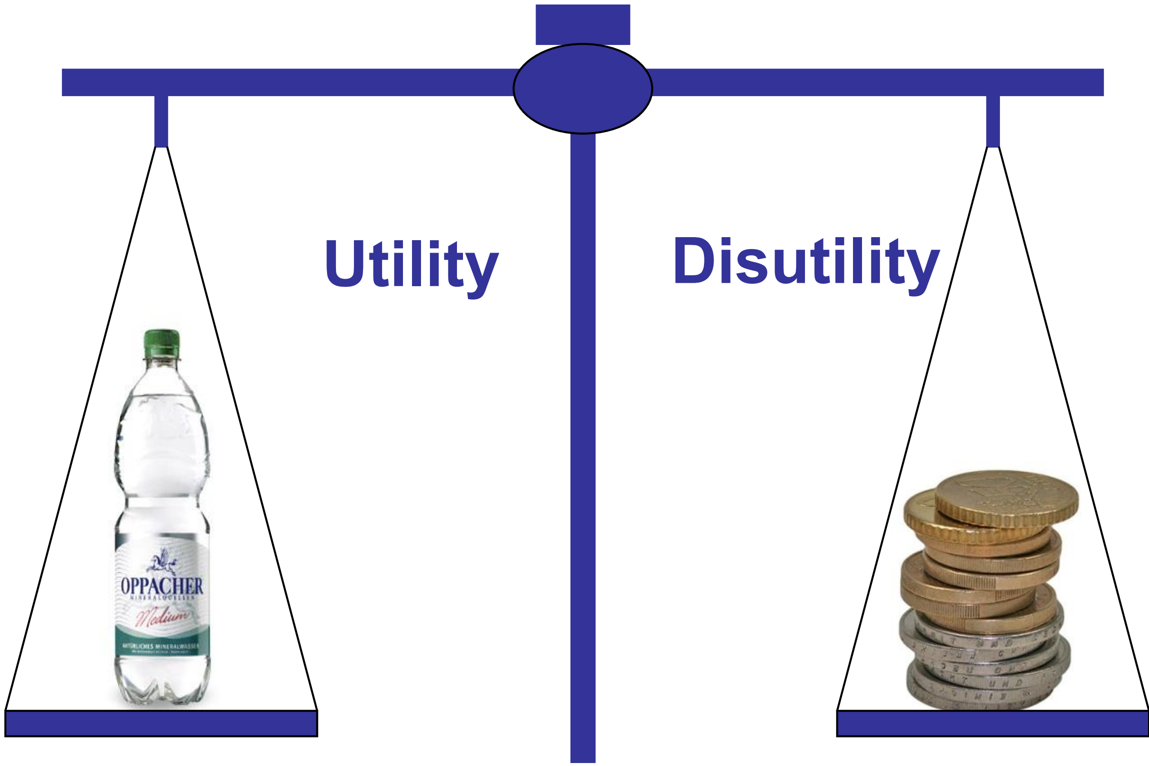
Concept development and testing



Conjoint analysis – Product attributes and features

Aims of preference measurement

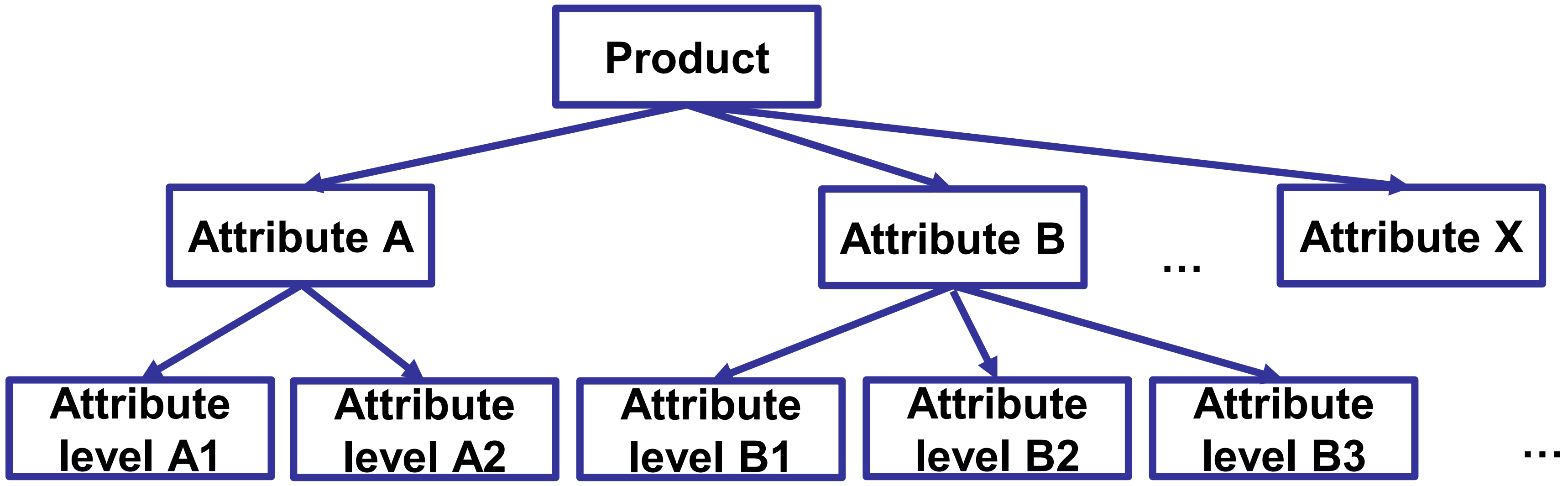
- Determine optimal products as combinations of attractive attribute levels
- Measure the relative utility value of attribute levels and their contribution to the overall utility value of a product
- Predict market shares
- Identify market segments
- Identify market opportunities/ potentials
- Price response functions



Preference models and assumptions

Assumptions:

- The overall utility of a product can be divided into the utility values of the product attributes/components



- Utility function captures the evaluation of the attribute levels.
- Deviations from the ideal attribute level lead to a lower overall utility value of a product.

Practical advices for determining the relevant attributes and attribute levels

Approaches to determine attributes and levels

Approaches not based on a pre-study

- Researchers or managers' instincts (Phantasialand)
- Advertising leaflets, websites, product reviews

Approaches based on qualitative pre-studies

- Process tracking
- Personal interviews
- Focus groups

Approaches based on quantitative pre-studies

- Direct ratings of attribute importances
- Repertory-grid technique (identify similarities and dissimilarities)

→ Open issue: How many attributes should be included?

The idea of conjoint-analytic preference measurement

Advantages:

- Respondents evaluate whole products. →Realistic decision situation
- Experimental setup: Flexibility in designing and presenting attributes and attribute levels
- Pressure to make a trade-off decision

Disadvantages:

- Should only be used for simple products with only a few attributes
- Laborious
- Risk of **information overload**

The idea of conjoint-analytic preference measurement

Video: Conjoint

https://www.youtube.com/watch?v=Su2qIr
Tmv1c&t=2s

Choice-based conjoint analysis

Rational of the approach:

- Let respondents choose between product concepts, because this is what consumers normally do in the marketplace. Respondents make decisions in a series of choice tasks.

Comparison to compositional preference measurement:

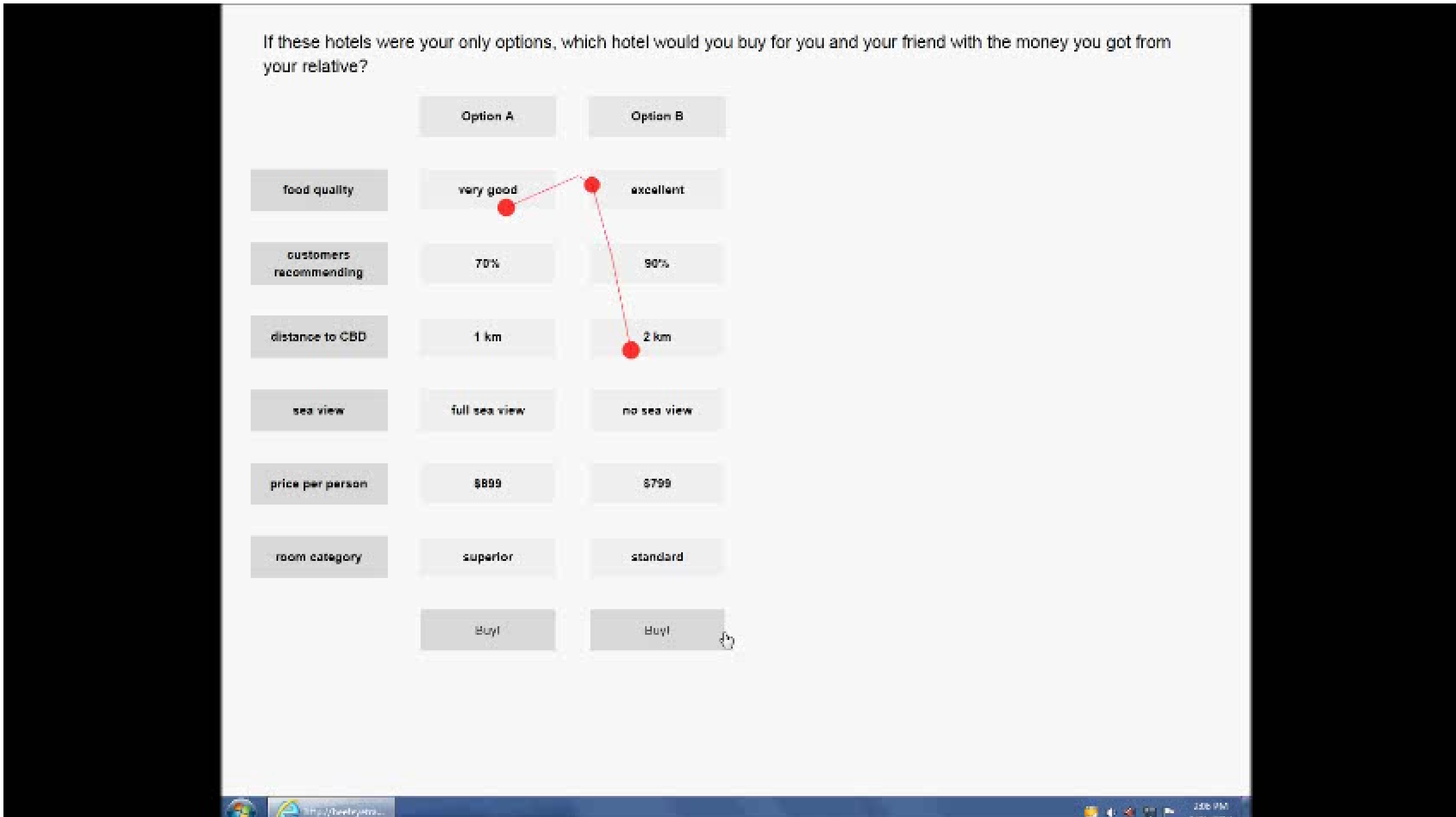
- Most realistic form of preference measurement
- A no-choice parameter can be estimated capturing low purchase intention
- Flexibility in the design of choice stimuli (shelf designs)
- New and better ways to incentive-align respondents?

Choice-based Conjoint analysis is also known as discrete choice modelling.

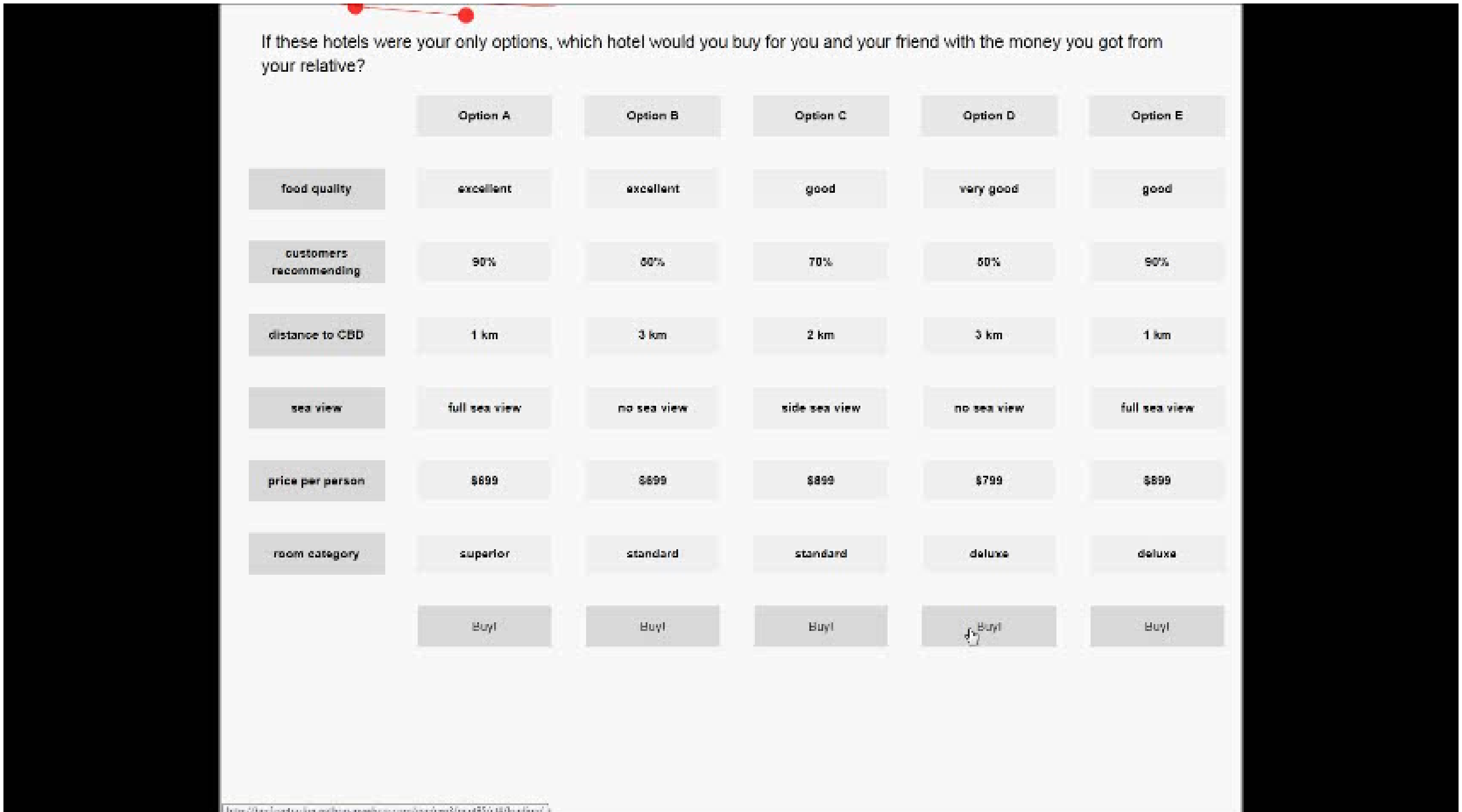
Choice-based conjoint analysis – Example 1 (virtual shelf)

Choice-based conjoint analysis – Example 2 (virtual shelf)

Choice-based conjoint analysis – Information processing in pairs and quints



Choice-based conjoint analysis – Information processing in pairs and quints



If these hotels were your only options, which hotel would you buy for you and your friend with the money you got from your relative?

	Option A	Option B	Option C	Option D	Option E
food quality	excellent	excellent	good	very good	good
customers recommending	90%	60%	70%	50%	90%
distance to CBD	1 km	3 km	2 km	3 km	1 km
sea view	full sea view	no sea view	side sea view	no sea view	full sea view
price per person	$899	$699	$899	$799	$899
room category	superior	standard	standard	deluxe	deluxe
	Buy!	Buy!	Buy!	Buy!	Buy!

Conjoint analysis

Interactive CBC illustration

www.sawtoothsoftware.com/baseball

Find more videos here:

http://www.sawtoothsoftware.com/support/videos

Contact details

Martin Meißner
Professor of Digital Marketing

Technical University of Munich
TUM School of Management
Bildungscampus 9
74076 Heilbronn
Tel. +49 7131 2641 8840

office.cdt@mgt.tum.de
www.wi.tum.de